

6	(a)	Magnetic flux density is the <u>force acting per unit current per unit length</u> on a conductor placed at <u>right angles to a magnetic field</u>	B1
	(b)(i)	The particle has a velocity component (v_y) that is perpendicular to the B-field and another component (v_x) parallel to the B-field. Due to <u>v_y always being perpendicular to the B-field</u>, there is <u>magnetic force</u> that is <u>always perpendicular to v_y</u>, hence resulting in <u>circular motion</u> of the particle. Due to <u>v_x always being parallel to the B-field</u>, there is <u>no magnetic force</u>, hence resulting in <u>linear motion</u> of the particle. Combining the two motions, the resulting path is helical.	B1 B1
	(ii)	The component of the velocity involves in the circular motion is $v_y = v \sin \theta$. The magnetic force acting on the charge acts as its centripetal force. $Bqv_y = \frac{mv_y^2}{r}$ $r = \frac{mv_y}{Bq} = \frac{mv \sin \theta}{Bq}$	B1 B1 M1 A0
	(iii)	$v \sin \theta = r\omega$ Since $r = \frac{mv \sin \theta}{Bq}$ $v \sin \theta = \frac{mv \sin \theta}{Bq} \times \frac{2\pi}{T}$ $T = \frac{2\pi m}{Bq}$ Since the expression for period T does not contain θ, T is independent of θ.	M1 A1

7	(c)	The magnitude of the induced e.m.f. is directly proportional to the rate of	B1
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